
SOFTWARE REQUIREMENTS SPECIFICATION

for

CAREER CRAFT

Version 1.0

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Project Guide

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1 Introduction

1.1 Purpose

The purpose of this project is to design and develop "Career Craft", an AI-powered web application aimed at preparing college students, particularly engineering students, for campus placements. This application integrates technical, aptitude, and soft skills training through features like mock interviews, resume analysis, aptitude tests, coding challenges, group discussion simulations, and personalized learning roadmaps. The primary goal is to provide an affordable, accessible, and adaptive platform that leverages AI/NLP/ML techniques to enhance student readiness, ensuring success in competitive job markets with real-time insights and scalable solutions.

1.2 Document Conventions

This document is structured to comprehensively outline the Career Craft project, detailing its training modules, AI-driven features, system architecture, and development considerations. Key conventions include:

- **UI:** Refers to the web-based user interface displayed for student interaction.
- **User Interaction:** Encompasses student inputs such as quizzes, resume uploads, and interview responses using the application.

Intended Audience and Reading Suggestions: This SRS targets the development team, project stakeholders (e.g., instructors, college administration), and evaluators. Readers should review the document in sequence to grasp the full scope, starting with the introduction for context, followed by technical and feature-specific sections.

1.3 Product Scope

Career Craft is an AI-powered web application designed to deliver a comprehensive placement preparation experience for engineering students. The scope includes mock interview simulations, resume building with ATS optimization, adaptive technical and aptitude quizzes, soft skills heatmaps, skill gap analysis, coding challenges, group discussion simulators, company knowledge provision, and domain-specific preparation. The project aims to utilize open-source AI/NLP/ML tools to provide cost-effective, scalable training solutions, deployable locally or on free cloud platforms, with optional enhancements, ensuring students are well-prepared for diverse industry roles.

1.4 References

[1]IEEE. IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society, 1998.

[2]Node.js Documentation: <https://nodejs.org/en/docs>

[3]React.js Documentation: <https://reactjs.org>

2 Overall Description

2.1 Product Perspective

Career Craft is a web-based application developed using Node.js, Express.js, and React.js, intended to bridge the gap between students and the competitive job market through AI-driven training. It integrates modules for resume analysis, mock interviews, aptitude tests, coding challenges, group discussions, and skill tracking to create a cohesive learning experience. The system comprises a Node.js backend with AI/ML integrations, a React.js frontend, and a MongoDB database to store student progress and performance data. This application stands out by offering personalized, real-time insights and adaptability, requiring only a standard web browser for access.

2.2 Product Functions

Career Craft's core functionality revolves around providing a holistic placement preparation experience. Key inputs and outputs include:

- **Inputs Collected from Students:**

- Resume uploads, quiz responses, coding submissions, and interview audio/video inputs.
- Student preferences (e.g., domain selection, difficulty levels).
- Optional login credentials for saving progress and analytics.

- **Outputs Provided to Students:**

- Real-time feedback on interviews, quizzes, and coding (e.g., clarity, confidence scores).
- Personalized learning roadmaps and skill gap reports with visualizations.
- Notifications for deadlines, quiz updates, and progress milestones.
- Adaptive quizzes, domain-specific questions, and company insights.

2.3 User Classes and Characteristics

The application supports three distinct user types:

- **Casual Users:** Students exploring the platform for basic preparation, focusing on quizzes and mock interviews without long-term commitment.
- **Registered Users:** Students who save progress, customize learning paths, and receive personalized feedback, enhancing their preparation.
- **Administrators/Developers:** Team members responsible for content updates, AI model training, bug fixes, and performance analytics, ensuring platform evolution.

2.4 Operating Environment

2.4.1 Database

A MongoDB database will store student data (e.g., progress, scores, skill assessments) with secure access for analytics and updates. As a NoSQL database, MongoDB provides flexibility in handling unstructured data, such as user-generated content from quizzes and interviews, ensuring efficient data management in both local and cloud setups.

2.4.2 Application

The application runs as a web application, accessible via modern browsers, requiring no specialized hardware beyond a standard computer with internet connectivity. It is designed to be cross-platform, supporting operating systems like Windows, macOS, and Linux, with optional deployment on free cloud platforms for enhanced scalability.

2.5 Design and Implementation Constraints

2.5.1 Database

The database must support efficient queries for real-time analytics and be scalable to handle increasing student data with future expansions. It should adhere to open-source standards to maintain cost-effectiveness and allow integration with AI/ML models without proprietary dependencies.

2.5.2 Application

The application must be optimized for web performance, ensuring compatibility with major browsers. Security measures must protect student data, and the design must prioritize intuitive UI/UX for diverse technical backgrounds. Implementation will leverage open-source tools to promote affordability and ease of maintenance.

2.6 User Documentation

Documentation will include setup guides, in-app tutorials on using AI features, and developer notes on Node.js, Express.js, and React.js integration. Additionally, it will provide API references for backend services and troubleshooting sections for common issues encountered during usage.

2.7 Assumptions and Dependencies

- Assumes compatibility with modern web browsers (e.g., Chrome, Firefox).
- Relies on stable internet connectivity for real-time features and cloud deployment.
- Assumes users have basic computer literacy and access to peripherals like microphones and webcams for interactive modules such as mock interviews.

3 External Interface Requirements

3.1 User Interfaces

- A clean, responsive web layout with clear navigation (e.g., dashboards, progress trackers) to guide students through modules.
- Intuitive controls for uploading resumes, recording interviews, submitting code, and accessing reports, ensuring ease of use.
- Key information (e.g., skill gaps, notifications) displayed prominently, accessible within 10 seconds of login.
- Consistent aesthetic (e.g., professional theme) with synchronized audio feedback for mock interviews and discussions.

3.2 Hardware Interfaces

- Standard computer with a microphone and webcam for mock interviews and group discussions.
- No specialized hardware required beyond a stable internet connection.

3.3 Software Interfaces

- **Name:** Node.js
Version: Latest stable release
Source: nodejs.org
Purpose: Backend runtime for server-side logic and API development.
- **Name:** Express.js
Version: Latest
Source: expressjs.com
Purpose: Web framework for building APIs.
- **Name:** React.js
Version: Latest
Source: reactjs.org
Purpose: Frontend framework for dynamic user interface.

3.4 Communications Interfaces

- Internet Protocol (IP) for real-time notifications, cloud sync, and data transmission.
- Secure data transmission for student progress using HTTPS.

4 System Features

4.1 Mock Interview Simulator

Career Craft's mock interview simulator uses AI-driven tools (spaCy, BERT, Whisper) to provide a realistic interview experience. Key aspects include:

- **Speech-to-Text:** Converts audio inputs to text for analysis.
- **Feedback:** Real-time feedback on clarity, confidence, and domain-specific responses.
- **Customizable Questions:** Adaptive question banks tailored to selected domains (e.g., AI, web development).
- **Performance Tracking:** Scores and suggestions for improvement based on speech and content analysis.

This feature ensures students practice effectively, receiving instant, objective feedback to enhance interview skills.

4.2 Resume Builder and Analyzer

The resume module leverages NLP (spaCy, sentence-transformers) for comprehensive resume management. Key elements include:

- **Parsing:** Extracts skills, experience, and education from uploaded resumes.
- **Job Matching:** Uses cosine similarity to align resumes with job requirements and ATS optimization.
- **Feedback:** Provides personalized suggestions for improvement and gap identification.
- **ATS Optimization:** Ensures resumes meet modern applicant tracking system standards.

This module helps students create competitive resumes tailored to their target roles.

4.3 Typing Speed Improver

A gamified module enhances typing skills with adaptive exercises. Key aspects include:

- **Adaptive Exercises:** ML-based coding challenges that adjust difficulty based on performance.
- **Progress Tracking:** Monitors speed and accuracy with real-time feedback.
- **Gamification:** Rewards and levels to motivate consistent practice.

This feature equips students with the skills needed for technical assessments, emphasizing speed and precision in typing.

4.4 Technical Quiz Platform

The technical quiz module offers adaptive learning with analytics. Key elements include:

- **Adaptive Quizzes:** IRT-based quizzes on DSA, databases with difficulty scaling.
- **Analytics:** Visualizations (Chart.js) of strengths, weaknesses, and progress trends.

This ensures targeted technical skill development and readiness for evaluations.

4.5 Aptitude Test Simulator

This module provides adaptive tests for non-technical skills. Key aspects include:

- **Adaptive Tests:** Quizzes on math, logical reasoning, and verbal skills with varying difficulty.
- **Feedback:** Instant scoring and suggestions for improvement.

This helps students build foundational aptitude required for placement tests.

4.6 Coding Challenge Platform

The coding module offers timed problems for practice. Key elements include:

- **Timed Challenges:** Coding problems with real-time feedback and auto-grading.
- **Adaptive Difficulty:** ML-based adjustment based on user performance.

This prepares students for coding rounds in interviews.

4.7 Group Discussion Simulator

This feature simulates group discussions with AI participants. Key aspects include:

- **AI-Driven Simulation:** Feedback on participation, clarity, and leadership.
- **Topic Variety:** Diverse topics with real-time analysis.

This enhances soft skills for group discussion rounds.

4.8 Soft Skills Heatmap/Confidence Tracker

This module assesses soft skills through sentiment and speech analysis. Key aspects include:

- **Sentiment Analysis:** Uses VADER and TextBlob to evaluate responses.
- **Speech Analysis:** Provides feedback on tone and confidence from interview recordings.
- **Heatmaps:** Visual representations of soft skill strengths (e.g., communication, time-management) and areas for improvement.

This helps students enhance interpersonal skills critical for placements.

4.9 AI-Driven Skill Gap Analyzer

The analyzer compares skills with job requirements. Key elements include:

- **Comparison:** Analyzes data from quizzes, resumes, and interviews against target jobs.
- **Recommendations:** Suggests resources and roadmaps to address gaps.

This provides targeted guidance for skill improvement.

4.10 Domain Selection and Preparation

This module tailors preparation to user-selected domains. Key aspects include:

- **Tailored Content:** Quizzes, roadmaps, and interview questions (BERT-based) for domains like AI, web development.
- **Personalization:** Adaptive paths based on user input and performance.

This ensures domain-specific readiness.

4.11 Company Knowledge Provider

This feature offers insights on companies. Key elements include:

- **Curated Insights:** Web scraping and NLP for hiring processes and news.
- **Updates:** Real-time company-specific preparation tips.

This helps students prepare for targeted companies.

4.12 Real-Time Notification Provider

The notification system keeps students updated using Socket.IO. Key elements include:

- **Alerts:** Notifications for quiz updates, interview deadlines, and progress milestones.
- **Push Notifications:** Real-time updates on company news and QA activity.
- **Scheduling:** Automated reminders for task completion.

This ensures students stay on track with their preparation timeline.

4.13 Student Analyzer

The analyzer tracks overall performance with ML-based insights. Key aspects include:

- **Performance Tracking:** Uses scikit-learn to analyze quizzes, interviews, and soft skills.
- **Visualizations:** Chart.js-based reports on strengths, weaknesses, and trends.
- **Progress Reports:** Detailed feedback for continuous improvement.

This provides a holistic view of student readiness for placements.

4.14 Usecase Diagram

The use case diagram for the AI-Powered Placement Preparation and Personal Growth Web Application illustrates a comprehensive ecosystem designed to empower college students, particularly engineering students, in navigating competitive campus placements through an integrated suite of AI-driven tools. At its core, the system encompasses key actors such as students interacting with functionalities like the Mock Interview Simulator, Resume Builder, Typing Speed Improver, Technical Quiz Platform, Soft Skills Heatmap/Confidence Tracker, AI-Driven Skill Gap Analyzer, Real-Time Notification Provider. Implemented with a robust stack including FastAPI backend, React.js frontend, MongoDB/PostgreSQL database, and open-source ML libraries this modular design ensures scalability, affordability via local or cloud deployment.

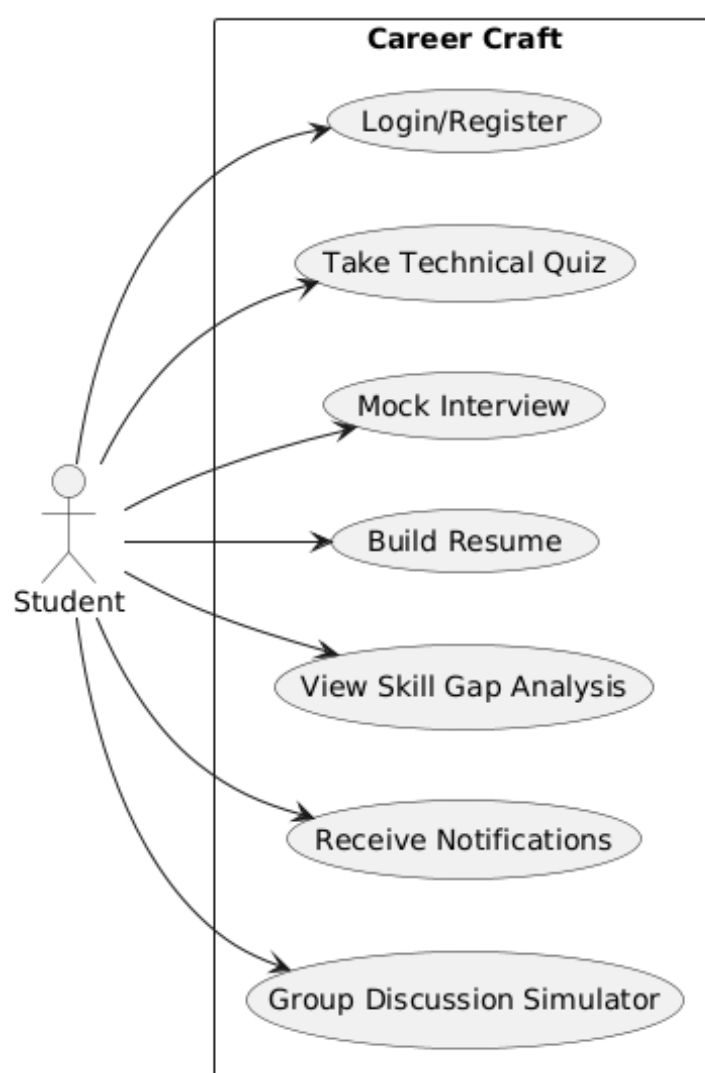


Figure 5.2: Usecase Diagram of Career Craft

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- Application must load within 5 seconds on standard hardware.
- Maintain responsive UI with minimal latency for real-time features.
- Database queries completed within 1 second for analytics.
- Optimize resources to fit within 4 GB RAM constraints.
- Support at least 100 concurrent users without degradation in response time.

5.2 Safety Requirements

- Ensure secure handling of student data to prevent breaches.
- Include warnings for session breaks (e.g., every 60 minutes) to avoid fatigue.
- Avoid complex inputs that could lead to user errors.
- Implement error handling to prevent application crashes and data loss.

5.3 Security Requirements

- Encrypt student data (e.g., resumes, progress) with JWT and bcrypt.
- Restrict database access to authorized developers via authentication.
- Protect against data tampering with input validation.
- Comply with data protection regulations such as GDPR for user privacy.

5.4 Software Quality Attributes

- **Availability:** Platform operational 99.9
- **Usability:** Intuitive interface learnable within 15 minutes.
- **Maintainability:** Modular code for easy updates and scalability.
- **Reliability:** System should recover gracefully from failures with minimal downtime.

6 Other Requirements

6.1 Privacy Policies

Career Craft prioritizes user privacy with a robust policy aligned with global data protection standards. Key elements include:

- **Data Transparency:** Clearly outlines collected data (e.g., resumes, quiz results), its usage (e.g., performance analytics), and storage locations (e.g., secure servers).
- **Consent Management:** Offers an opt-in process during registration, with options to disable non-critical data tracking.
- **User Control:** Provides students with tools to access, modify, or remove their personal data through a dedicated settings panel.

This approach fosters trust and ensures responsible data stewardship.

6.2 Collection of Personal Information

Personal data collection is kept minimal and purpose-driven to support platform functionality:

- **Optional Identification:** Students can opt for a username for progress tracking or remain anonymous.
- **Performance Metrics:** Collects only essential data (e.g., test scores, skill assessments) for personalized feedback, stored with high security.

These practices maintain a balance between usability and user privacy.

6.3 Support and Maintenance

Career Craft includes provisions for ongoing support and system maintenance to ensure reliability:

- **Regular Updates:** Scheduled maintenance to enhance features, fix bugs, and improve performance based on user feedback.
- **Backup Systems:** Implements automated backups to prevent data loss, with recovery protocols for unexpected failures.

This ensures the platform remains functional and user-friendly over time.

Appendix A: Glossary

- **AI:** Artificial Intelligence – Technology enabling adaptive learning and analysis.
- **NLP:** Natural Language Processing – Techniques for text and speech analysis.
- **ATS:** Applicant Tracking System – Software used by employers to screen resumes.
- **ML:** Machine Learning – Algorithms for performance tracking and recommendations.